

Experiment No. 6

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Experiment Title

Use string functions and regex for email validation, data extraction, and text analysis.

Software / Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript

Experiment Program Code

Files Used:

- regex.html
- regex.css
- regex.js

regex.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Quantum Data Regex Extractor</title>
  <link rel="stylesheet" href="regex.css">
</head>
<body>

  <div class="matrix-bg"></div>

  <div class="container">
    <header>
      <div class="system-status">SYS_STATUS: ONLINE // SECURE_NODE</div>
      <h1>Data Regex Extractor</h1>
      <p class="subtitle">Extract, validate, and transform data streams via Regular Expressions</p>
    </header>

    <!-- Input Area -->
    <div class="input-section">
      <label for="rawText"><span class="terminal-char">&gt;</span> Input Raw Text Stream:</label>
      <textarea id="rawText" placeholder="Student Name: John Doe"
PRN: 202611049218
Email Address: john.doe@university.edu
Phone Number: 9876543210
Department: Computer Science">Student Name: John Doe
PRN: 202611049218
Email Address: john.doe@university.edu
Phone Number: 9876543210
Department: Computer Science</textarea>
      <button id="processBtn">Execute Parsing Core</button>
    </div>

    <!-- Output Area -->
    <div class="output-section hidden" id="outputWrapper">

      <div class="card">
        <h2>// Extracted Data Matrices</h2>
        <div class="grid">
          <div class="label">[01] Student Name:</div>
          <div class="value text-glow" id="outName">-</div>

          <div class="label">[02] PRN String:</div>
          <div class="value text-glow" id="outPrn">-</div>

          <div class="label">[03] Email Endpoint:</div>
          <div class="value">
            <span id="outEmail" class="text-glow">-</span>
            <span id="emailStatus" class="badge">-</span>
          </div>

          <div class="label">[04] Phone Comm Line:</div>
          <div class="value">
```

```

        <span id="outPhone" class="text-glow">-</span>
        <span id="phoneStatus" class="badge">-</span>
    </div>

    <div class="label">[05] Department Hub:</div>
    <div class="value text-glow" id="outDept">-</div>
</div>
</div>

<div class="card">
    <h2>// Stream Telemetry Statistics</h2>
    <div class="grid mini">
        <div class="label">Total Word Blocks:</div>
        <div class="value font-bold neon-cyan" id="statWords">0</div>

        <div class="label">Total Data Bits (Chars No Spaces):</div>
        <div class="value font-bold neon-cyan" id="statChars">0</div>
    </div>
</div>

<div class="card">
    <h2>// Lowercase Mutated Output</h2>
    <textarea id="outLowercase" readonly></textarea>
</div>

</div>
</div>

<script src="regex.js"></script>
</body>
</html>

```

regex.css

```
:root {
  --bg-dark: #030712;
  --panel-bg: rgba(17, 24, 39, 0.85);
  --border-neon: #06b6d4;
  --border-neon-dim: rgba(6, 182, 212, 0.2);
  --primary-glow: #3b82f6;
  --text-neon: #a5f3fc;
  --text-muted: #94a3b8;

  --success-neon: #10b981;
  --success-bg: rgba(16, 185, 129, 0.15);
  --error-neon: #ef4444;
  --error-bg: rgba(239, 68, 68, 0.15);
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: 'Courier New', Courier, monospace;
  background-color: var(--bg-dark);
  color: #e2e8f0;
  padding: 40px 20px;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: flex-start;
  position: relative;
  overflow-x: hidden;
}

/* Sci-fi mesh background overlay */
.matrix-bg {
  position: fixed;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  z-index: -1;
  background-image:
    linear-gradient(rgba(6, 182, 212, 0.03) 1px, transparent 1px),
    linear-gradient(90deg, rgba(6, 182, 212, 0.03) 1px, transparent 1px);
  background-size: 20px 20px;
}

.container {
  width: 100%;
  max-width: 850px;
  background: var(--panel-bg);
  border: 1px solid var(--border-neon-dim);
  border-radius: 12px;
  padding: 32px;
  box-shadow: 0 0 30px rgba(6, 182, 212, 0.15), inset 0 0 15px rgba(59, 130, 246, 0.1);
  backdrop-filter: blur(12px);
}

.system-status {
  font-size: 0.75rem;
  color: var(--border-neon);
  text-shadow: 0 0 8px rgba(6, 182, 212, 0.6);
  letter-spacing: 2px;
  margin-bottom: 0.5rem;
  text-align: center;
}

h1 {
  font-size: 2.2rem;
  margin-bottom: 0.5rem;
  color: #fff;
  text-shadow: 0 0 15px rgba(255, 255, 255, 0.2);
  text-align: center;
  text-transform: uppercase;
  letter-spacing: 1px;
}

.subtitle {
  color: var(--text-muted);
  text-align: center;
  margin-bottom: 2.5rem;
  font-size: 0.95rem;
}

.input-section {
  margin-bottom: 2rem;
}

label {
  font-weight: 600;
  display: block;
  margin-bottom: 0.75rem;
  color: var(--text-neon);
  text-shadow: 0 0 5px rgba(165, 243, 252, 0.4);
}

terminal-char {
```

```

    color: var(--border-neon);
}

/* FIX: Text areas configured strictly for horizontal safety locking */
textarea {
  width: 100%;
  max-width: 100%;
  min-width: 100%;
  height: 200px;
  min-height: 120px;
  max-height: 500px;
  padding: 16px;
  background-color: rgba(3, 7, 18, 0.9);
  border: 1px solid var(--border-neon-dim);
  border-radius: 6px;
  color: #38bdf8;
  font-family: inherit;
  font-size: 1rem;
  line-height: 1.6;
  resize: vertical; /* Up/down resize tracking boundary lock */
  box-shadow: inset 0 0 10px rgba(0, 0, 0, 0.8);
  transition: all 0.3s ease;
}

textarea:focus {
  outline: none;
  border-color: var(--border-neon);
  box-shadow: 0 0 15px rgba(6, 182, 212, 0.3), inset 0 0 5px rgba(6, 182, 212, 0.2);
}

button {
  background: linear-gradient(90deg, #0891b2, #2563eb);
  color: white;
  border: 1px solid rgba(255, 255, 255, 0.1);
  padding: 14px 24px;
  font-size: 1.1rem;
  font-weight: 600;
  font-family: inherit;
  border-radius: 6px;
  cursor: pointer;
  margin-top: 1.25rem;
  width: 100%;
  text-transform: uppercase;
  letter-spacing: 1px;
  box-shadow: 0 0 15px rgba(37, 99, 235, 0.3);
  transition: all 0.2s ease;
}

button:hover {
  box-shadow: 0 0 25px rgba(6, 182, 212, 0.6);
  filter: brightness(1.1);
}

button:active {
  transform: scale(0.99);
}

.hidden {
  display: none !important;
}

.card {
  background: rgba(10, 15, 30, 0.7);
  border: 1px solid var(--border-neon-dim);
  border-radius: 8px;
  padding: 24px;
  margin-top: 1.5rem;
}

.card h2 {
  font-size: 1.1rem;
  color: var(--border-neon);
  text-shadow: 0 0 8px rgba(6, 182, 212, 0.4);
  margin-top: 0;
  margin-bottom: 1.5rem;
  border-bottom: 1px solid var(--border-neon-dim);
  padding-bottom: 0.75rem;
  text-transform: uppercase;
  letter-spacing: 1px;
}

.grid {
  display: grid;
  grid-template-columns: 200px 1fr;
  gap: 16px;
  align-items: center;
}

.grid.mini {
  grid-template-columns: 320px 1fr;
}

.label {
  color: var(--text-muted);
  font-size: 0.9rem;
}

.value {
  word-break: break-all;
  font-size: 1.05rem;
}

```

```

    color: #f8fafc;
}

.text-glow {
  color: #e0f2fe;
  text-shadow: 0 0 4px rgba(224, 242, 254, 0.3);
}

.font-bold {
  font-weight: 700;
}

.neon-cyan {
  color: #22d3ee;
  text-shadow: 0 0 8px rgba(34, 211, 238, 0.5);
}

.badge {
  display: inline-block;
  padding: 3px 8px;
  font-size: 0.75rem;
  font-weight: 700;
  border-radius: 4px;
  margin-left: 10px;
  letter-spacing: 1px;
  text-transform: uppercase;
  vertical-align: middle;
}

.badge.valid {
  background-color: var(--success-bg);
  color: var(--success-neon);
  border: 1px solid var(--success-neon);
  box-shadow: 0 0 8px rgba(16, 185, 129, 0.3);
}

.badge.invalid {
  background-color: var(--error-bg);
  color: var(--error-neon);
  border: 1px solid var(--error-neon);
  box-shadow: 0 0 8px rgba(239, 68, 68, 0.3);
}

#outLowercase {
  height: 140px;
  background-color: rgba(3, 7, 18, 0.6);
  color: #64748b;
  border-style: dashed;
}

```

regex.js

```
document.getElementById('processBtn').addEventListener('click', function() {
    const rawText = document.getElementById('rawText').value;

    // --- 1. Extraction Regex Patterns ---
    const nameMatch = rawText.match(/Student Name:\s*(.*)/i);
    const prnMatch = rawText.match(/PRN:\s*(.*)/i);
    const emailMatch = rawText.match(/Email Address:\s*([\w\.-]+\@[\w\.-]+\.\w+)/i);
    const phoneMatch = rawText.match(/Phone Number:\s*(\d+)/i);
    const deptMatch = rawText.match(/Department:\s*(.*)/i);

    // Fallbacks if field does not exist
    let extractedName = nameMatch ? nameMatch[1].trim() : "Not Found";
    let extractedPrn = prnMatch ? prnMatch[1].trim() : "Not Found";
    let extractedEmail = emailMatch ? emailMatch[1].trim() : "";
    let extractedPhone = phoneMatch ? phoneMatch[1].trim() : "";
    let extractedDept = deptMatch ? deptMatch[1].trim() : "Not Found";

    // --- 2. Data Validations ---
    const emailRegex = /^[a-zA-Z0-9._%+-]+\@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
    const isEmailValid = emailRegex.test(extractedEmail);

    const phoneRegex = /^\d{10}$/;
    const isPhoneValid = phoneRegex.test(extractedPhone);

    // --- 3. Text Statistics ---
    const totalWords = rawText.trim() === "" ? 0 : rawText.trim().split(/\s+/).length;
    const totalCharsNoSpaces = rawText.replace(/\s+/g, '').length;

    // --- 4. Replacements & Transformations ---
    const lowercaseText = rawText.toLowerCase();

    if (extractedDept.toLowerCase() === "computer science") {
        extractedDept = "Information Technology";
    }

    // --- 5. DOM Injection ---
    document.getElementById('outName').textContent = extractedName;
    document.getElementById('outPrn').textContent = extractedPrn;
    document.getElementById('outEmail').textContent = extractedEmail || "Not Found";
    document.getElementById('outPhone').textContent = extractedPhone || "Not Found";
    document.getElementById('outDept').textContent = extractedDept;

    // Email Badge Update
    const emailBadge = document.getElementById('emailStatus');
    if (emailBadge) {
        emailBadge.className = `badge ${isEmailValid ? 'valid' : 'invalid'}`;
        emailBadge.textContent = isEmailValid ? 'Valid' : 'Invalid';
    }

    // Phone Badge Update
    const phoneBadge = document.getElementById('phoneStatus');
    if (phoneBadge) {
        phoneBadge.className = `badge ${isPhoneValid ? 'valid' : 'invalid'}`;
        phoneBadge.textContent = isPhoneValid ? 'Valid' : 'Invalid';
    }

    // Stats injection
    document.getElementById('statWords').textContent = totalWords;
    document.getElementById('statChars').textContent = totalCharsNoSpaces;

    // Lowercase text box injection
    document.getElementById('outLowercase').value = lowercaseText;

    // Reveal output area
    document.getElementById('outputWrapper').classList.remove('hidden');
});
```

Application Description

The Data Regex Extractor accepts raw text containing student details. JavaScript uses regular expressions to extract the student name, PRN, email address, phone number, and department. It validates email and phone data, calculates text statistics, converts the input to lowercase, and displays the processed information dynamically.

Features Implemented

Feature	Description
Regex Extraction	Extracts student name, PRN, email, phone number, and department.
Email Validation	Checks whether the extracted email follows a valid email pattern.
Phone Validation	Checks whether the phone number contains exactly 10 digits.
Text Statistics	Counts words and characters after removing spaces.
String Transformation	Converts the complete input text into lowercase.

Feature	Description
Data Replacement	Changes Computer Science department output to Information Technology.
DOM Manipulation	Displays extracted data, validation badges, and statistics dynamically.

Output

DATA REGEX EXTRACTOR

SYS_STATUS: ONLINE // SECURE_NODE

> Input Raw Text Stream:

Student Name: John Doe
PRN: 202611049218
Email Address: john.doe@university.edu
Phone Number: 9876543210
Department: Computer Science

EXECUTE PARSING CORE

// EXTRACTED DATA MATRICES

[01] Student Name:	John Doe
[02] PRN String:	202611049218
[03] Email Endpoint:	john.doe@university.edu VALID
[04] Phone Comm Line:	9876543210 VALID
[05] Department Hub:	Information Technology

// STREAM TELEMETRY STATISTICS

Total Word Blocks: 14

Total Data Bits (Chars No Spaces): 104

Output Summary

- Accepts raw text containing student information.
- Extracts required fields using regular expressions.
- Validates email and phone number formats.
- Displays word count and character count without spaces.
- Converts the input text into lowercase.
- Dynamically displays the extracted information and validation status.

Result / Conclusion

The Data Regex Extractor was implemented successfully using HTML, CSS, and JavaScript. The experiment demonstrated regular expressions, string functions, text analysis, data extraction, validation, string transformation, and DOM manipulation.